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AIRBOX HOUSING OF THE JUNGFRAU DETECTOR FOR IN VACUUM X-RAY DIAGNOSTICS

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Abstract

At the European XFEL facility, ultrabright X-ray radiation is employed to investigate phenomena in a variety of sample materials with the highest spatial and temporal resolution. For X-ray detection, the scientific 'Jungfrau' detector is frequently employed, as it matches the parameters of the provided X-ray beams. Originally developed by the Paul Scherrer Institute for in-air use at the SwissFEL, a detector housing has been designed and constructed at European XFEL to meet the requirements of in-vacuum operation at the scientific instruments for high-energy density physics (HED) and material-induced dynamics (MID).

The in-vacuum version of the Jungfrau detector is applied in various specialized diagnostics and methods aimed at resolving atomic lattice structures through X-ray diffraction, observing laser-induced microscopic material changes such as shock-wave dynamics via X-ray imaging and small-angle X-ray scattering, or probing plasma temperatures with inelastic X-ray spectroscopy. To exploit the coherence of the X-rays, the design includes a windowless X-ray photon beam path extending from the source to the sample and detector plane.

This contribution presents the housing design for a single module and showcases fully integrated solutions for selected X-ray diagnostics, incorporating multiple modules to enhance functionality.

DETECTOR HOUSING

The Jungfrau camera [1] is a compact hybrid-pixel detector with a 512 x 1024 sensor format featuring auto-gain switching for high dynamic range measurements and single photon sensitivity. Furthermore, it can record up to 16 frames with currently up to 300 kHz acquisition rate which renders the device well-suited for the X-ray pulse delivery pattern provided by the European XFEL facility to its scientific instruments (Fig. 1).

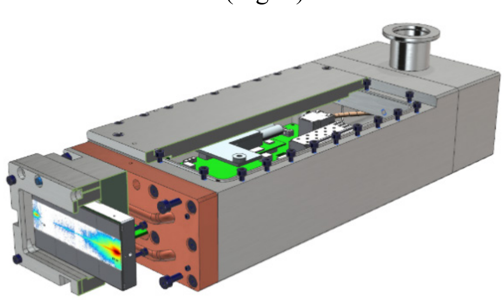

Figure 1: Assembled Jungfrau detector inside the airbox housing (partially cut view).

While other beamlines operate their detectors in an ambient environment, at the HED [2] and MID [3] instruments they must be integrated into the experimental setups accommodated within large vacuum chambers which requires the use of a customized housing. The detector housing assembly, depicted in Fig. 1, satisfies the following development goals: a) window-less sensor directly exposed to vacuum in order to maximize quantum efficiency, minimize scattering background from an entrance window and maintain the coherent properties if the X-ray beam; b) compact and slim housing dimensions to enable the flexible integration into multiple diagnostic setups; c) a nearly gapless modular assembly of two detector housings to double the sensor format to either a 512 x 2048 or 1024 x 1024 pixel matrix, respectively.

Technical Description

The detector key parts consist of a sensor module, referred to as front-end module (FEM), and the printed circuit board (PCB) which features the read-out electronics. While the FEM is designed to operate in vacuum, the PCB must remain at atmospheric condition to allow for convective cooling of the electric components.

The housing design is illustrated via an explosion drawing in Fig 2. It makes use of a vacuum barrier flange attached onto the PCB, mounted against the housing from the inside.

The backend adapter flange allows to mount a standard DN25-KF bellow which accommodates power, data, and trigger cables and support media (water, air/nitrogen). Different version types of this flange point the bellow into different directions. Depending on the vacuum chamber size and the location of the designated chamber feedthrough port the bellow length can be adapted.

Cooling of the detector must be provided to the FEM and to the PCB read-out electronics. Cooling of the PCB is realized via a flow of nitrogen introduced into the airbox housing via a small tube with the bellow acting as a gas outlet. Stable operation of the FEM sensor however requires a temperature stability below 100 mK (milli-degree Celsius). Therefore, a water-cooled system driven by a refrigerator unit is used, which provides sufficient flow to a copper cooling block containing brazed-in copper tubes.

The FEM is assembled to an interface block which keeps the system modular and enables rapid replacement of the sensor during experiments (in case of X-ray damage). Interfaces that are connecting the sensor to the cooling block are contacted with graphite foils (Fig. 2). A sensor cap aims to protect the sensor but also allows to mount experiment specific filter foils in order to block optical light (black Kapton) or attenuate the X-ray signal (thin metal foils).

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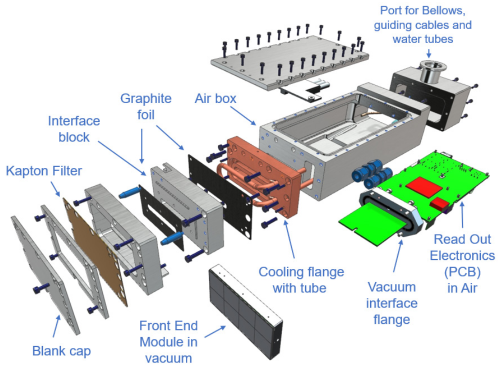

Figure 2: Explosion drawing of the air box housing for a single Jungfrau module detector. The key components are labelled.

APPLICATIONS AT HED

At the HED instrument the in-vacuum version of the Jungfrau detector is used inside the large interaction chamber. Here, high-energy and high-intensity laser beams focused onto samples create transient plasmas of high energy density and high temperatures which are diagnosed using a variety of techniques. Some key diagnostic methods are briefly discussed in the following.

HAPG von Hámos Spectrometer

The spectrometer [4] is designed to operate primarily between 5 and 12 keV using Highly Annealed Pyrolitic Graphite (HAPG) crystals to disperse the X-rays, scattered or emitted from the sample, onto the detector in cylindrical von Hámos geometry. From the measured spectra conclusions can be drawn about the plasma conditions, e.g. about the sample temperature. As illustrated in Fig. 3, the crystal is located half distance between the Jungfrau detector and the source and is offset from the source-detector line by its radius-of-curvature. The working distance for a given photon energy is determined by the crystal Bragg angle and is experimentally set by the source to detector distance. The spectrometer assembly is motorized and can be adjusted relative to the sample in angle and in translation, whereas the relative position of the crystal to the Jungfrau detector is set manually using a long holder arm (Fig. 3).

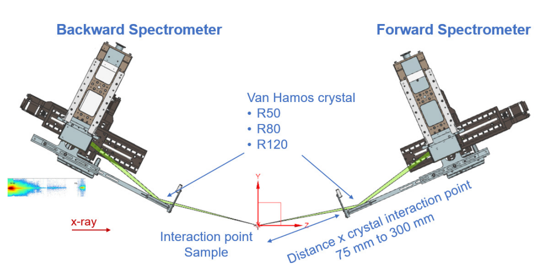

Figure 3: Schematic CAD view of the forward and backward spectrometer in von Hámos geometry.

One Megapixel Configuration for X-ray Diffraction

The JF is mostly used as a single module inside the experimental chamber of HED, positioned at selected angles (azimuthal, polar) around the sample interaction point for X-ray diffraction measurements. From the angularly resolved scattering signal structural information of the sample can be resolved.

However, there is a minimum possible distance given by the sensor area and in some applications the active sensor area must be enlarged by combining two detector modules. Here, the airbox housing design allows the mounting of two modules with minimal gap size in such a way that two modules align either on the long sensor side, resulting in a square 1024 x 1024 pixel sensor, or on the short sensor side, resulting in an elongated 512 x 2048 pixel configuration. See Fig. 4 for a technical picture a) and respective example images from an experimental campaign using the square b) and c) elongated configuration.

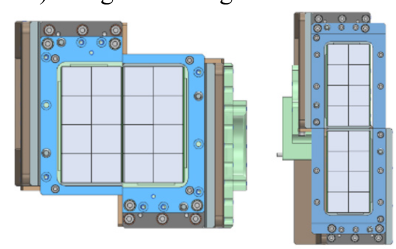
Figure 4: Technical drawing of a combined two module assembly resembling a squared and elongated sensor configuration.

Diced Crystal Analyser (DCA) Spectrometer

In order to measure scattering signal with a spectral resolution on a level of a few millielectronvolt, which gives insight into atomic motion dynamics, a silicon diced crystal analyser is used [5]. The X-ray optic collects, disperses and re-focuses the scattered light from a sample onto a detector which located directly above the sample plane. Therefore, a single Jungfrau module has been integrated into a motorized assembly structure enabling the fine alignment of this diagnostic. Figure 5 illustrates the setup in a CAD model of the HED interaction chamber with three DCAs located in backscattering geometry (facing the sample from the rear side). The Jungfrau detector, sits above the sample in the chamber centre and can be translated in three dimensions.

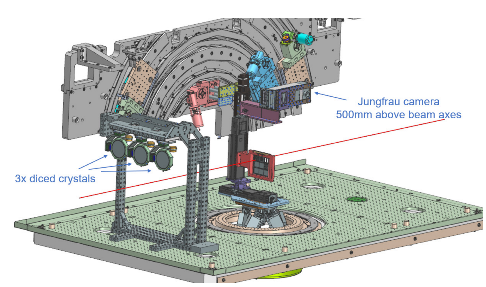

Figure 5: CAD image of the diced crystal analyser (DCA) setup with a single Jungfrau module located above the sample with translation capabilities.

APPLICATIONS AT MID

The MID instrument is used to explore the structure and dynamics of matter on nm to subatomic length scales and

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us-to-fs time scales. This is achieved through a variety of experimental techniques such as X-ray Photon Correlation Spectroscopy (XPCS), Coherent Diffraction Imaging (CDI), X-ray holography, Ptychography and Ultrafast X-ray Diffraction (UXRD). Two separate single-module Jungfrau detectors are used at MID to complement the large primary AGIPD detector [6]. The instrument originally could be setup in three different geometrical configurations: Wide Angle X-ray Scattering (WAXS), Small Angle X-ray Scattering (SAXS) and a Large Field of View (LFV). Small-area Jungfrau detectors mounted inside a dedicated vacuum chamber (called Multiple-Detector Stage, MDS) [7] enable Ultra-Small Angle X-ray Scattering (USAXS) geometry for the benefit of studying samples on um-to-nm length scales or simultaneous SAXS/WAXS geometries to study structure and dynamics of samples on both, atomic and nanometre length scales in the same experiment.

The design of the Jungfrau detector housing allows the use of both Jungfrau detectors ether in vacuum inside the dedicated MDS vacuum chamber or in-air at a user-defined position of the experimental set-up.

Ultras Small Angle X-ray Scattering

The important feature of the multiple detector stage at MID is that it can be placed either on the existing AGIPD detector arm or as a standalone device on a separate girder (Fig. 6).

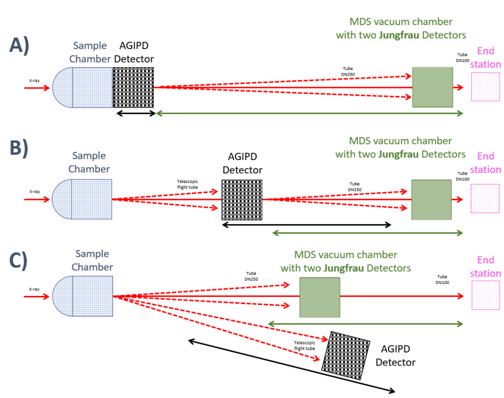

Figure 6: Schematic view of the MID instrumental set up: A) Large field of view and MDS at 7m; B) AGIPD at 3m with MDS at 7m; C) AGIPD in WAXS geometry with MDS on its own girder to cover the SAXS region.

When mounted on the AGIPD arm behind the primary detector in SAXS geometry of the instrument, the distance from the sample to the primary detector can still be varied in the range from 3 to 6 meters, while allowing capturing X-rays scattered on very small angles by the Jungfrau detector.

Figure 7 shows how the sensitive areas of AGIPD and two Jungfrau detectors overlap when AGIPD is placed at 3 m from the sample and JF is at 7 m distance from the sample. Blue rectangle depicts a point projection of the AGIPD central "hole" opening from the sample on to the Jungfrau sensor planes.

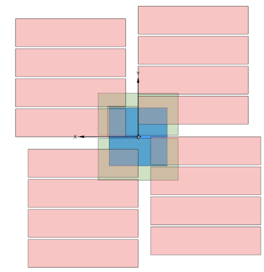

Figure 7: Parallel projection view of AGIPD (red) and two Jungfrau (green) detector sensor planes. Blue rectangle showing the overlap of the point projection of the AGIPD central hole opening from the sample onto the Jungfrau sensor planes.

This enables performing XPCS, CDI, inline X-ray holography and ptychography experiments with the same windowless setup.

OUTLOOK

In the current version of the detector housing the PCB with the read-out electronics is attached straight in line behind the FEM such that the peripheric read-out electronics extending normal to the active sensor area. For many experimental configurations, however, the X-ray detection is required in the horizontal plane where the housing competes with other setup relevant components such as laser optics that lay in the same plane. Utilizing the third dimension for the detector body is advantageous in order to provide valuable space in the scattering plane. In order to achieve this a new version of the housing is planned where the PCB extends parallel to the sensor area. A first design study is shown in Fig. 8 left. The PCB is encapsulated in the green box orients vertically while the sensor active area faces horizontally towards the sample. Furthermore, the cooling circuit is planned to be outside the air-box separated from the electronics board, to avoid damage of the electronics in case of a water leakage. A prototype of a 3D-printed cooling block made of aluminium has been manufactured to verify vacuum compatibility (Fig. 8 right).

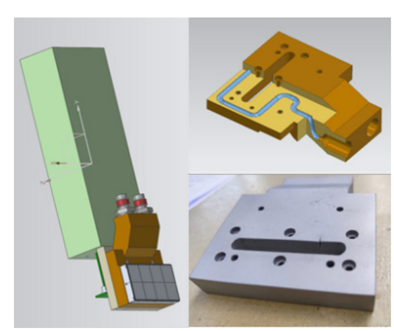

Figure 8: (Left) Design study of the planned Jungfrau air-box housing with orthogonally aligned PCB board. (Right) Prototype of a 3D printed cooling block with integrated water channel.

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